

Weekly Update: Dynamic DAG Prediction for Proactive ILP Execution

1 Project Motivation

The objective of this project is to reduce the execution latency of agent workflows that contain optimization tasks solved by Integer Linear Programming (ILP). An agent workflow can be represented as a dynamically evolving directed acyclic graph (DAG). Unlike a static workflow, the complete DAG may not be known before execution begins because new nodes and dependencies can be generated from intermediate tool outputs and agent decisions.

Normally, an ILP task cannot start until its corresponding workflow node and required inputs have been generated. This creates an opportunity for speculative execution. If the system can predict that an ILP-related node is likely to appear in the near future, it can prepare or start the ILP computation early. A correct prediction allows ILP execution to overlap with earlier workflow operations and may reduce waiting time on the critical path.

The intended process is therefore to observe the current partial DAG, predict the node types that may appear within a limited future depth, and use sufficiently confident predictions to trigger ILP computation. The current work focuses on evaluating whether future workflow nodes are predictable enough to support this mechanism.

2 Prediction Formulation

At the current stage, each workflow is represented as the sequence in which its nodes become visible or execute. Given an observed node history

$$x_1, x_2, \dots, x_t, \tag{1}$$

the predictor estimates the next H node types,

$$x_{t+1}, x_{t+2}, \dots, x_{t+H}, \tag{2}$$

where H is a limited prediction horizon. I currently evaluate horizons of one, two, and three nodes. The semantic node taxonomy includes `inspect`, `edit`, `shell`, `test`, `reason`, `fetch`, and `STOP`.

This formulation does not attempt to reconstruct the complete future DAG. Instead, it provides short-range lookahead that may be sufficient to identify an upcoming ILP-related node and begin computation before the node reaches the execution frontier.

3 Dataset and Experimental Setup

The experiments use approximately 10,000 SWE-Trace/OpenHands workflow traces. The dataset contains about 596,745 nodes, with an average of 58.7 actions and a median of 54 actions per workflow. The test split contains 2,000 workflows and 116,677 prediction samples.

The node distribution is highly imbalanced. The `inspect` type occurs 318,352 times and represents approximately 53% of all nodes, while some types are rare. Consequently, overall accuracy is influenced heavily by common node types. Macro-F1 is also reported to give equal weight to each semantic class.

Two prediction methods were evaluated:

- **N-gram predictor:** predicts future nodes using local transition statistics from the most recent node types. Context windows of 0, 1, 2, 3, 5, and 8 were evaluated. Window 0 is the context-free baseline.
- **GRU predictor:** learns a recurrent representation of the observed workflow history. History lengths of 1, 2, 3, 5, 8, and 16 were evaluated. The model uses a 32-dimensional embedding, a hidden size of 64, dropout of 0.1, and class-weighted training.

Both methods were evaluated with prediction horizons of 1, 2, and 3, a top- k value of 4, and a confidence threshold of 0.6 for speculative-execution gating.

4 Evaluation Metrics

The experiments use the following metrics:

- **Next-node accuracy:** whether the first predicted node is correct.
- **Macro-F1:** the unweighted mean of the per-class F1 scores.
- **Exact-sequence accuracy:** whether all H predicted nodes are correct and in the correct order.
- **Top-4 coverage:** whether the true future sequence appears among the four candidate predictions.
- **Gate trigger rate:** the fraction of prediction points at which confidence reaches the threshold of 0.6.
- **Gate precision:** the fraction of triggered predictions for which the predicted future sequence is exactly correct.

Gate precision and trigger rate are particularly relevant to proactive ILP execution. High precision limits wasted ILP work, while a useful trigger rate is needed to create enough opportunities for latency reduction.

5 Results

Table 1 presents representative results from the strongest overall configurations. The best N-gram configuration uses a context window of 5, while the GRU configuration uses a history length of 16.

H	Model	Configuration	Next Acc.	Macro-F1	Exact Seq.	Gate Prec.
1	N-gram	Window 5	58.41%	29.56%	58.41%	80.81%
1	GRU	History 16	56.35%	37.44%	56.35%	79.02%
2	N-gram	Window 5	57.88%	28.32%	39.42%	76.17%
2	GRU	History 16	56.38%	37.48%	35.04%	79.97%
3	N-gram	Window 5	57.42%	28.99%	29.44%	65.09%
3	GRU	History 16	56.13%	37.39%	25.54%	77.84%

Table 1: Representative N-gram and GRU prediction results.

The context-free exact-sequence baselines are 52.14%, 35.96%, and 27.32% for horizons 1, 2, and 3, respectively.

5.1 N-gram Results

The N-gram predictor performs strongly despite its simplicity. A context window of 5 gives the highest exact-sequence accuracy at all three horizons: 58.41%, 39.42%, and 29.44%. These values improve over the context-free baselines by 6.28, 3.46, and 2.13 percentage points, respectively. The results indicate that agent workflows contain useful short-range transition patterns.

Increasing the window from 5 to 8 reduces performance. For example, horizon-1 accuracy decreases from 58.41% to 56.73%. This likely reflects N-gram sparsity: long exact contexts occur less frequently and therefore have less reliable probability estimates. With the current dataset size, a five-node context appears to provide the best balance between useful history and statistical support.

5.2 GRU Results

The GRU produces a different trade-off. Increasing history length from 1 to 16 raises horizon-1 Macro-F1 from 25.89% to 37.44%, while accuracy changes from 54.41% to 56.35%. Thus, longer histories mainly improve predictions for less frequent node types rather than substantially improving overall accuracy.

The current GRU does not outperform the context-free baseline on multi-step exact sequences. Its best horizon-2 and horizon-3 exact-sequence accuracies are 35.04% and 25.54%, compared with baselines of 35.96% and 27.32%.

Possible causes include autoregressive error propagation, the mismatch between the class-weighted training objective and sequence-level accuracy, and the limited input representation. The model currently observes only node types and does not include DAG topology, tool arguments, task content, or intermediate execution results.

Nevertheless, the GRU remains useful as a high-confidence predictor. With a history length of 16, gate precision remains between approximately 78% and 80% across the three horizons. The corresponding trigger rates are 38.96%, 24.29%, and 18.93%. Therefore, the GRU can identify a smaller subset of relatively safe multi-step predictions even though its overall exact-sequence accuracy is limited.

6 Implications for Proactive ILP Execution

The results support the feasibility of using workflow prediction to trigger ILP execution, especially at short horizons. N-gram with window 5 is currently the strongest general predictor because it achieves the highest complete-sequence accuracy and has very low inference overhead. GRU with history length 16 is more balanced across node classes and may be helpful when ILP-related nodes are rare.

Prediction quality alone, however, does not determine whether speculation improves the system. The practical decision should be based on an expected-benefit model such as

$$\begin{aligned} \text{Expected Benefit} = & P(\text{correct trigger}) \times \text{saved latency} \\ & - P(\text{incorrect trigger}) \times \text{wasted computation} \\ & - \text{prediction overhead.} \end{aligned} \tag{3}$$

If an incorrect ILP launch is expensive, the system should use a conservative high-precision gate. If spare compute resources are available, a lower threshold may increase coverage and produce more opportunities to hide ILP latency.

It may also be unnecessary to predict every future node exactly. The operational goal is to determine whether an ILP-relevant node will appear and whether enough information is available to begin its computation. A target-specific prediction objective may therefore be more useful than full sequence reconstruction.

7 Current Conclusions and Next Steps

The experiments demonstrate that short-term agent workflow evolution is predictable. N-gram with a five-node window currently provides the best exact-sequence results and exceeds the context-free baseline at every tested horizon. GRU provides better Macro-F1 and reliable confidence-filtered predictions, but its multi-step sequence prediction requires further improvement.

The next stage will connect these prediction results to actual ILP execution. The planned work is to identify ILP-relevant node types, report their class-specific precision and recall, calibrate confidence thresholds separately for each horizon, and measure both correct and incorrect speculative ILP launches. I will then compare end-to-end workflow latency with and without proactive ILP execution. Future GRU experiments will also incorporate richer workflow features, including DAG structure, task state, and node metadata.